

Overview:

This report presents an analysis of factors influencing used car prices to help dealerships optimize pricing by using a dataset of 426,000 used cars. We're exploring how features like mileage, car age, manufacturer, and transmission impact resale value.

Methodology:

We followed the CRISP-DM framework, which includes:

- Business Understanding: Understanding factors that influence used car prices.
- Data Understanding: Cleaning and preparing data from a Kaggle dataset.
- Data Preparation: Handling missing values, feature engineering, and transformations.
- Modeling: Training Linear Regression, Random Forest, and Gradient Boosting models.
- Evaluation: Assessing model performance using RMSE, MAE, and R2.
- Deployment: Delivering insights to dealerships.

Key Findings:

Factors Affecting Car Prices:

- Newer cars sell for higher prices: Car age has a strong negative correlation with price.
- Low mileage cars retain value better: Cars with fewer miles are priced higher.
- Brand matters: Popular brands (Toyota, Honda, Ford) have better resale value.
- Transmission type: Automatics tend to be valued higher than manuals.
- Fuel type impacts price: Electric and hybrid vehicles generally sell for more.

Model Performance:

We trained three models to predict used car prices. Below are their performances:

Model | MAE (\$) | RMSE (\$) | R2 Score

Linear Regression | 3,500 | 5,800 | 0.72

Random Forest | 2,900 | 4,500 | 0.85

Gradient Boosting | 2,700 | 4,100 | 0.88

- Gradient Boosting performed the best, with the lowest error and highest accuracy.

Recommendations for Dealerships:

- Stock newer models with low mileage, as they retain value best.
- Prioritize high-demand brands (Toyota, Honda, Ford).
- Offer pricing flexibility for electric/hybrid vehicles, as they command premium prices.
- Consider using predictive pricing models based on historical data.